

Cooking with a touch of science
and a dash of engineering

(Raspberry Pi controlled
sous-vide water bath)

Chris Swan

@cpswan

Sous-vide ([/suːˈviːd/](#); French for "under vacuum")

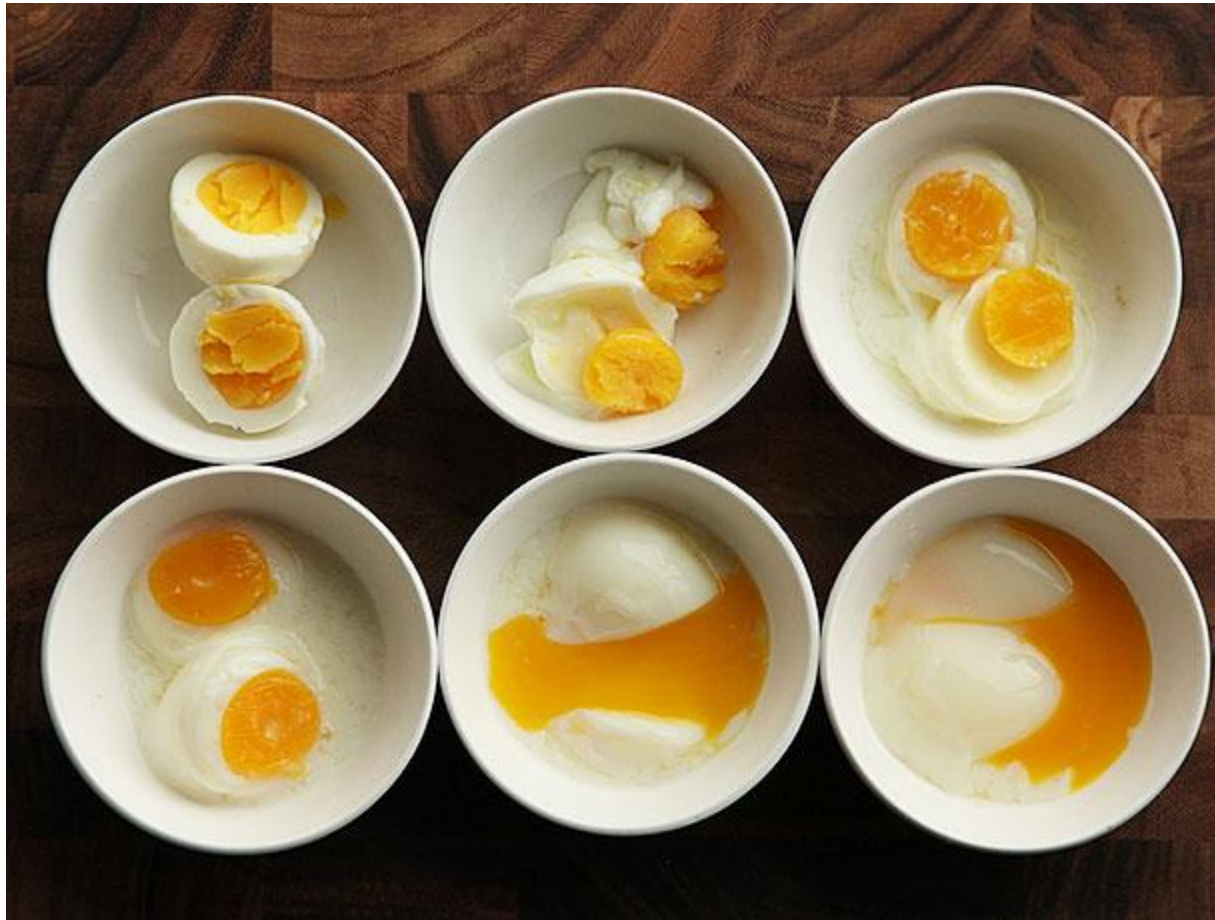


By BBCLCD - Own work, CC BY 3.0,
<https://commons.wikimedia.org/w/index.php?curid=32386182>



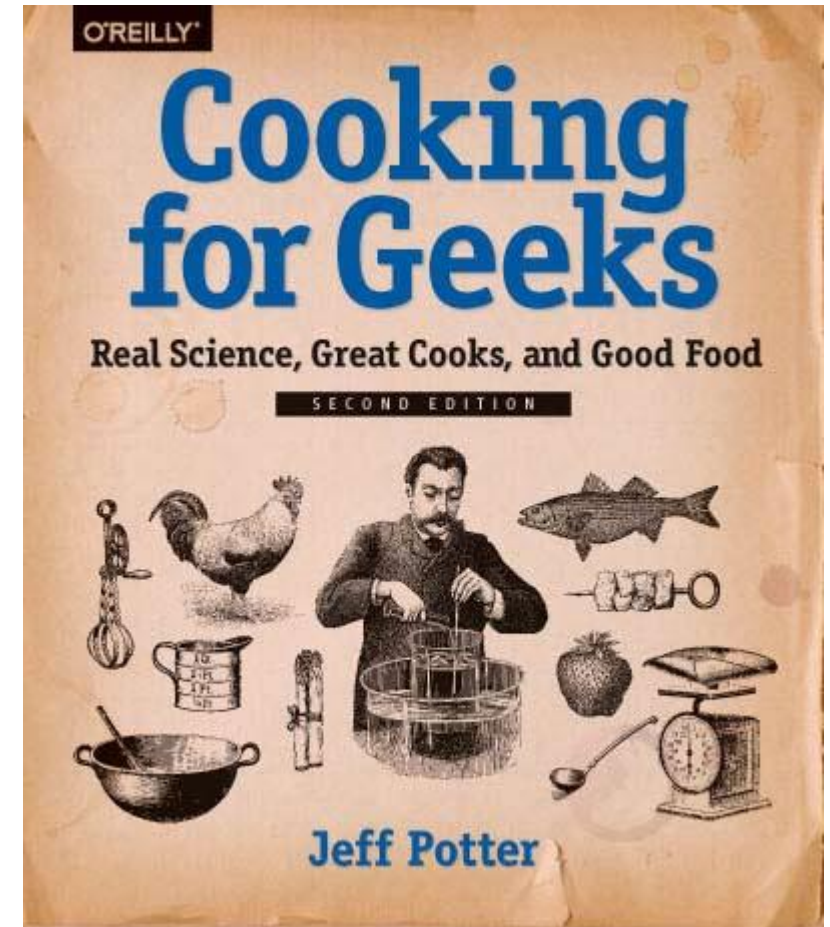
By JamieS93 - Template:Charlie the unicorn, CC BY-SA 3.0
<https://commons.wikimedia.org/w/index.php?curid=4904747>

Precise temperature control is the point (to denature just the right proteins)

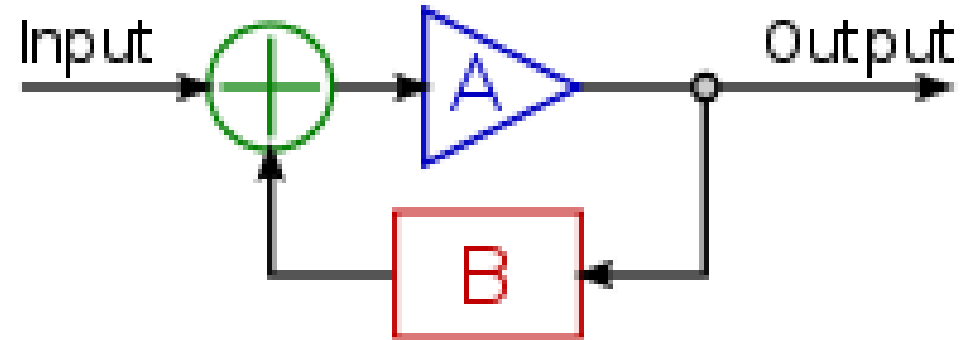
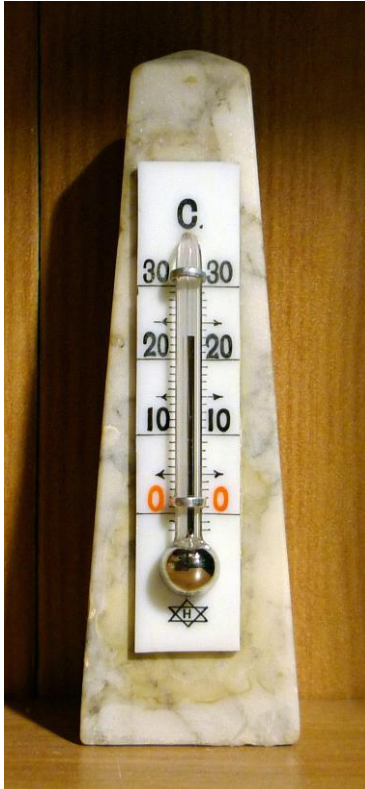


Photographs: J. Kenji Lopez-Alt

<https://www.seriouseats.com/2013/10/sous-vide-101-all-about-eggs.html>



We need a control system



By Intgr – Public Domain

https://en.wikipedia.org/wiki/Control_system#/media/File:Ideal_feedback_model.svg

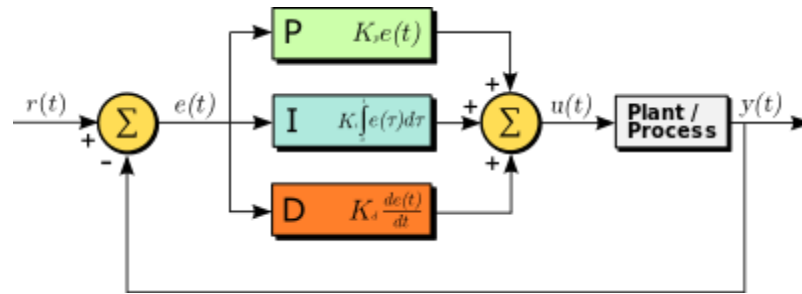
By Anonimski - Own work, CC BY-SA 3.0,

https://upload.wikimedia.org/wikipedia/commons/5/57/Mercury_Thermometer.jpg

By Acosta.eu - Own work, CC BY-SA 3.0,

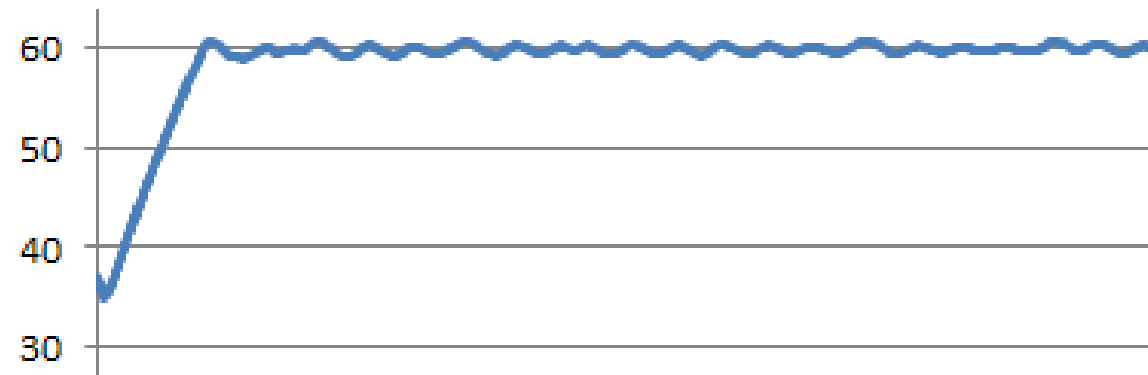
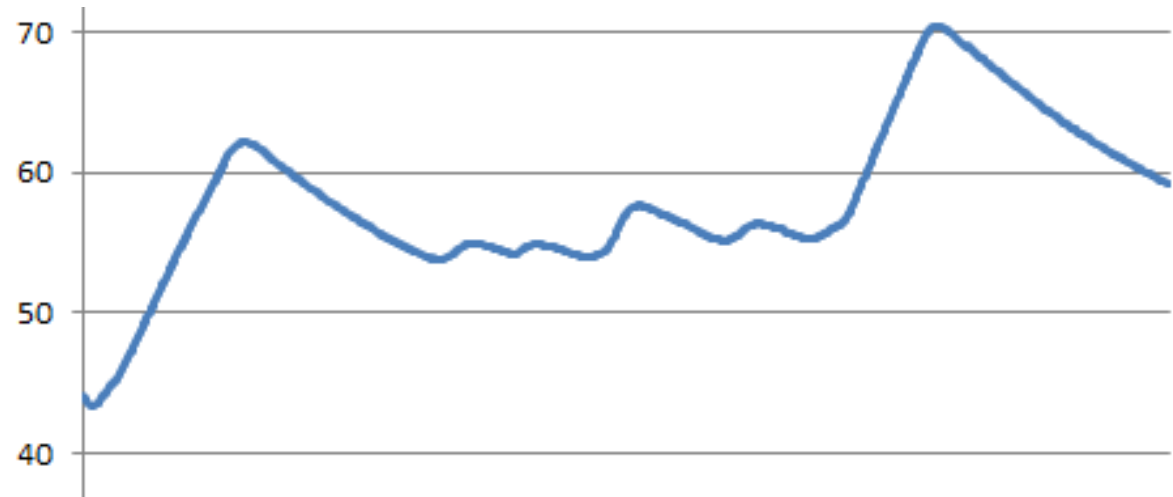
<https://upload.wikimedia.org/wikipedia/commons/d/d9/Heizstab-heatingelement-espressomachine.jpg>

DS18B20 – Raspberry Pi (Python PID) – 433MHz Remote Control Socket PWM – Slow Cooker



Control systems can be tricky

```
52 print("Entering control loop")
53 while True:
54     temperature=tempdata()
55     print(temperature)
56     error = target - temperature
57     intererror = intererror + error
58     power = B + ((P * error) + ((I * intererror)/100))/100
59     print power
60     if (power > 0):
61         pwr_tot = pwr_tot + power
62         pwr_ave = pwr_tot / pwr_cnt
63         pwr_cnt = pwr_cnt + 1
64         print pwr_ave
65     # Make sure that if we should be off then we are
66     if (state=="off"):
67         turn_off()
68     # Long duration pulse width modulation
69     for x in range (1, 100):
70         if (power > x):
71             if (state=="off"):
72                 state="on"
73                 print("On")
74                 turn_on()
75             else:
76                 if (state=="on"):
77                     state="off"
78                     print("Off")
79                     turn_off()
80     sleep(1)
```



Reverse engineering can also be tricky



```
time:      Sun Aug 26 15:31:59 2018
hardware:  433gpio
pulse:     2
rawlen:    65
pulselen:  205

Raw code:
2076 236 1373 480 1351 548 1267 573 1317 589 1237 1150 678 631 1251 574 1231 576
 1381 478 1299 1163 626 573 1287 578 1289 521 1349 1219 614 572 1361 471 1265 58
6 1301 1186 627 550 1443 402 1313 1226 590 610 1311 561 1281 532 1313 506 1344 5
26 1311 507 1341 501 1363 493 1408 438 1331 668 6992
--[RESULTS]--

time:      Sun Aug 26 15:31:59 2018
hardware:  433gpio
pulse:     6
rawlen:    66
pulselen:  207

Raw code:
1163 628 601 1238 593 1255 567 1287 572 1257 597 1268 1191 673 592 1260 593 1258
 566 1247 591 1260 1161 687 586 1267 574 1261 613 1252 1194 666 588 1246 566 127
8 573 1259 1186 671 580 1263 558 1293 1174 686 579 1254 602 1248 592 1254 561 12
61 594 1272 578 1270 590 1272 583 1299 573 1217 595 7055
--[RESULTS]--
```

Perfect brisket makes it all worthwhile 😊



Thanks for your time

I'm happy to answer questions

Loads more code, photos etc. at

<http://blog.thestateofme.com/2018/08/26/raspberry-pi-sous-vide-redux/>